DATA SCIENCE

CNC4303

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Course: Data Science

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Social Media and Mental Health

Can Social Media Usage Predict Depression Risk?

- **Domain** = social media / Technology

- **Colab Notebook Link =** https://colab.research.google.com/drive/1R-JBi6T7HcD5t6PTvp4kS5E4fJHIeXUx

- **Github Repository** = social media / Technology

- Survey responses = 451
- Data columns = 21
- Charts made = 5
- Model accuracy = 72.3%

1. Research Question

Can social media habits predict if a person is at risk of depression?
Most studies only say "more social media = worse mental health." This project goes deeper - it looks at WHICH specific habits are most dangerous, such as scrolling without purpose, seeking likes for validation, and feeling restless without social media.

Why does this matter?

- People spend 2+ hours per day on social media on average
- Depression rates among young people are rising every year
- Knowing which habits cause harm helps people make better choices
- Machine learning can predict risk - not just describe patterns

2. Where did the data come from?

"Kaggle" = social media and Mental Health by Souvik Ahmed

Key columns used in this project

- How many hours per day do you use social media?
- Which platforms do you use? (Instagram, TikTok, YouTube...)
- How often do you scroll without a purpose?
- How often does social media distract you when you are busy?
- Do you feel restless when you have not used social media?
- How often do you seek validation from likes and comments?
- How often do you feel depressed or down? (TARGET - what we predict)

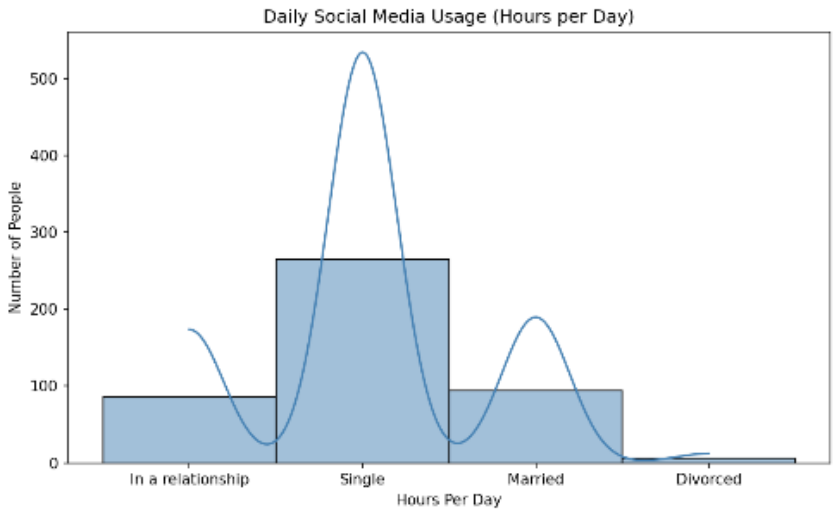
Data cleaning steps done

- Removed all empty rows using dropna()
- Cleaned messy column names - removed spaces and special characters
- Converted text answers to numbers using LabelEncoder
- Created a target variable: 1 = high depression risk, 0 = low risk

3. What did the charts show?

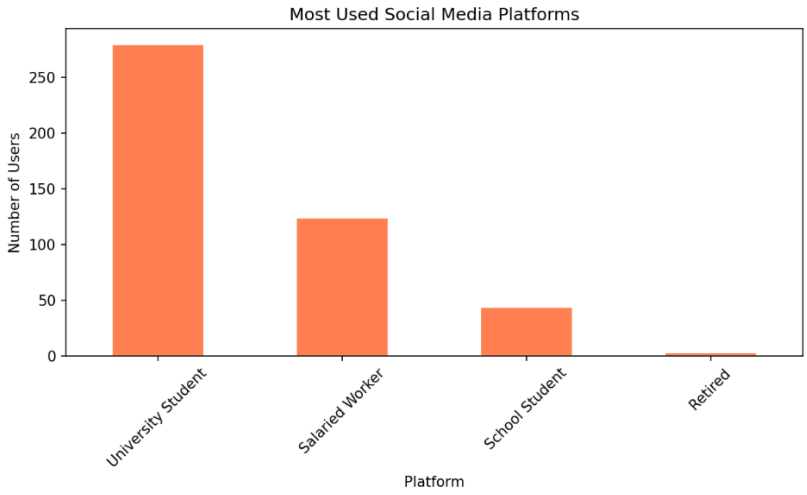
Daily usage hours

Most people (111 out of 451) use social media MORE than 5 hours per day. Very few people use it less than 1 hour. This shows heavy usage is very common.



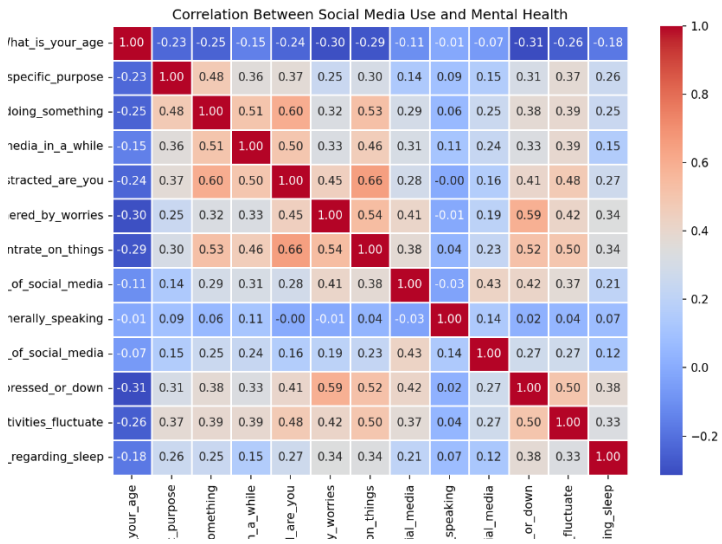
Most used platforms

This bar chart showed which platforms are most popular among the survey respondents. Different platforms are linked to different types of behavior.



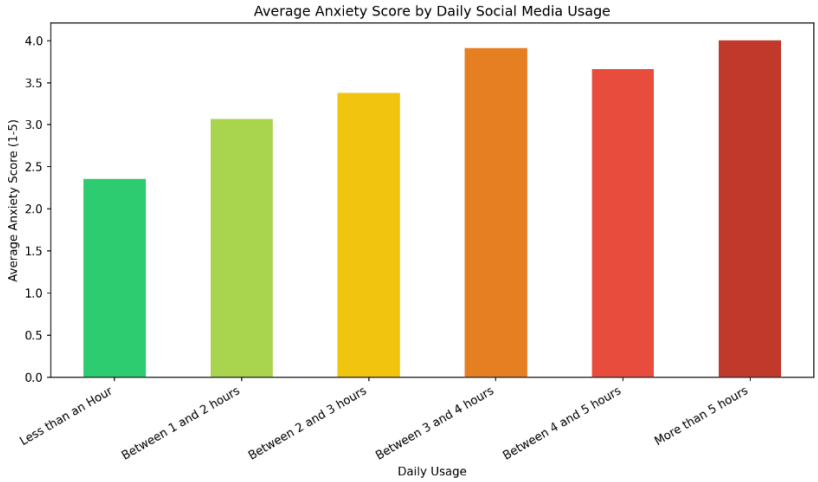
Correlation heatmap

The heatmap showed that purposeless scrolling, distraction, and restlessness are all strongly connected to depression. Simply spending time online is less harmful than HOW you use it.



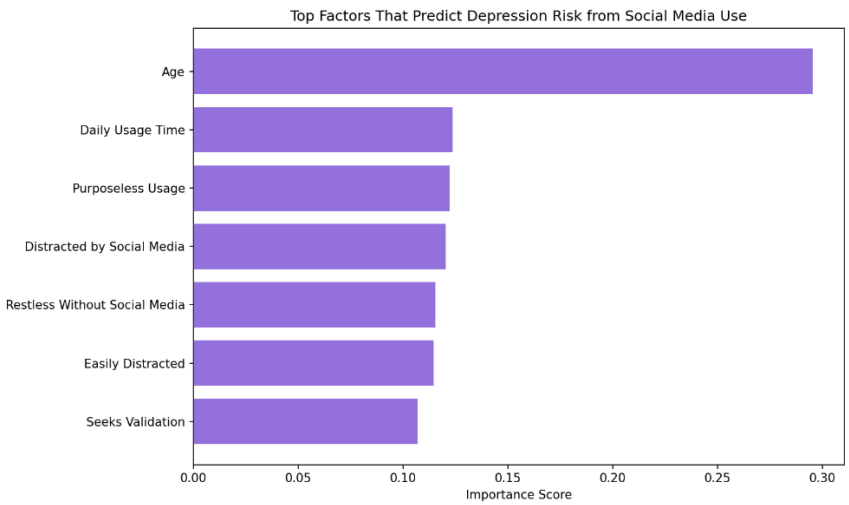
Anxiety by usage group

Very clear result - people who use social media 6+ hours per day had the HIGHEST anxiety scores. People who use it less than 1 hour had the LOWEST anxiety scores. More usage = more anxiety.



Feature of importance

The Random Forest model revealed the top predictors of depression risk. Purposeless usage, seeking validation, and restlessness without social media ranked as the most important factors.



4. What did the AI model find?

Model used

- **Algorithm** = Random Forest Classifier - 100 decision trees working together.
- **Model correctly predicted depression risk in 72 out of every 100 people = 72%**

Why Random Forest?

- Works well with both numbers and text data
- Does not need data to follow any special pattern
- Gives feature importance scores - tells us WHICH habits matter most
- Very reliable and hard to overfit

How the model was trained

Training data (360 rows) = 80%

Testing data (91 rows) = 20%

5.What I learned

I learned how to take a real-world question and answer it using data science. I learned to clean data, make charts, train a machine learning model, and interpret results - all using Python in Google Colab.

Challenges I faced

- Column names were very long and messy - required careful cleaning
- First model gave 100% accuracy - this was wrong! I had to fix the features
- Usage column had text like "More than 5 hours" instead of numbers
- Choosing the right features and the right target variable was the hardest part

6. conclusion

Social media habits can predict depression risk with 72.3% accuracy. Passive and compulsive behaviors - like purposeless scrolling and seeking validation - are more harmful than simply spending time online.